

The growing demand for accuracy, automation, and real-time monitoring in workforce management has led to the need for advanced employee tracking solutions. Traditional attendance systems such as manual registers, RFID swipes, and biometric devices often suffer from inefficiencies, human error, lack of scalability, and insufficient data transparency. These limitations become even more apparent in industrial or large-scale work environments where constant monitoring and accurate data logging are critical.

To address these challenges, we have developed a **BLE-Based Employee Tracking System**, a smart and efficient approach for automating employee attendance and calculating active work hours using **Bluetooth Low Energy (BLE)** technology. The primary goal of this project is to eliminate manual intervention and provide a reliable solution for attendance monitoring and work hour calculation.

The system operates by assigning each employee a BLE beacon—specifically the **SparkFun NanoBeacon IN100**—which emits a unique UUID signal continuously. These signals are detected by an **ESP32-based BLE receiver** placed at the entry and exit points of the workplace. Once detected, the system logs the precise entry and exit timestamps of each employee.

This data is then transmitted via **Wi-Fi** to a **cloud-based backend (Firebase)**, where it is processed to calculate the total number of active hours worked by each employee. The data is then displayed on a responsive **web dashboard**, accessible to managers and administrators for remote monitoring, analysis, and decision-making. The system is designed to be simple, energy-efficient, scalable, and easy to deploy in any office or industrial setup.

By minimizing human error, eliminating the need for physical contact, and allowing real-time data access, this solution not only streamlines attendance tracking but also significantly improves operational efficiency. Additionally, the system is cost-effective and eco-friendly, reducing the reliance on paper-based registers or complex biometric setups.

The **BLE-Based Employee Tracking System** serves as a practical example of how **IoT (Internet of Things)** technologies can be effectively applied to solve real-world administrative problems. It provides a foundation for future improvements, such as integration with payroll systems, employee notifications, or mobile-based alert systems. This project demonstrates the practical utility of combining hardware (ESP32, BLE beacons) with software (Firebase, web dashboard) to deliver an end-to-end automated solution for employee management.